AI IN STRUCTURAL HEALTH MONITORING FOR PREDICTIVE MAINTENANCE & SAFETY ASSURANCE

EXECUTIVE PROGRESS SUMMARY: PREVIOUS MDP, CURRENT IDP, HARDWARE UPGRADE & PROCUREMENT PLAN

BACSE291 — Innovative Design Project | Fall 2026 | Vellore Institute of Technology (VIT), Vellore
Guide: Dr. Senthil Kumar N (Emp ID: 15377, SCOPE) | **Aerial (15377IDP01):** Priyam S., Ayushman K. | **Ground (15377IDP02):** Nisargkumar J. (Lead), Aryan M., Mudit G.

1. PREVIOUS MDP PROJECT: PLY FORGE

• **Objective & Architecture:** Client-side WebGL 3D Point Cloud Processor in HTML5/ES6 without external libraries.
• **Implemented Capabilities:** Binary/ASCII PLY parsing (DataView/TextDecoder), 3D point cloud rendering, circle ROI filtering, 2.5D Delaunay/grid surface triangulation, and standardized PLY export.
• **Voxel Compression:** Spatial hashing downsampling point clouds by 60%–85% while preserving structural boundaries.
• **Cross-Project Reuse:** Provides the verified GPU rendering and digital twin engine for current IDP ground analytics. Eliminates software licensing costs.

2. CURRENT IDP WORK DEVELOPED & VERIFIED

• **Aerial Subsystem (15377IDP01):** Benewake TF-Luna 850nm ToF (<5g, 0.35W, 100 Hz) interfaced with ESP32. Firmware parses 9-byte packets, validates 0x59 0x59 sync and 8-bit checksum. Integrated median filter (N=5).
• **Two-Point Metrology Fit:** Universal straight-line model: $y = 1.0228x + 3.20$ cm (R^2=1.0000, RMSE=±0.02 cm across 39.0cm and 61.4cm physical benchmarks).
• **Ground Subsystem (15377IDP02):** Rolling baseline residuals ($\Delta d = d_{meas} - d_{ref}$); flags spalling cavities ($\Delta d \geq +3.5$cm) and bulges ($\Delta d \leq -3.5$cm).
• **3D Digital Twin:** Reconstructed 9,969-point room (.PLY) isolating Almari bulge (+65cm, Orange) and Gate recess (-60cm, Green). 5-tab Streamlit dashboard active.

3. HARDWARE UPGRADES & PROCUREMENT BUDGET

• **Engineering Need:** Manual sliding jitter (~2–3 cm) and sensor tilt elongation ($d_{true} = d_{meas} \cdot \cos\theta$) require dedicated hardware sensing and active airframe mobility.
• **YDLIDAR X2 360° LiDAR:** 3,000 pts/s, 7 Hz continuous planar slicing, 0.12–8.0m range, 3.3V UART (115200 bps), 5V 500mA. Ref price: ■**7,189.00**.
• **DFRobot Gravity Tilt Sensor:** 360° ball switch detecting pitch/roll to apply trigonometric compensation in ESP32 firmware. Ref price: ■**245.00**.
• **F450 Drone Kit + Pixhawk:** F450 frame, 4x 2212 motors, 4x 30A ESCs, Pixhawk 2.4.8 autopilot, 1045 props. Subtotal: ■**12,020.00**.
• **Power & Mounting:** 3S 4200mAh 35C LiPo (■2,450), iMAX B6 charger (■1,350), 5V 5A UBEC (■380), silicone damper & wiring (■740).
• **TOTAL BUDGET:** Essential: ■24,724 | Shipping: ■400 | Contingency (8%): ■2,000 | **TOTAL: ■27,124.00 INR** (MVP Option: ■8,214.00).

4. FUTURE ROADMAP & SYSTEM INTEGRATION

• **Phase 1–2 (Weeks 1–3):** Procure Priority 1 hardware (LiDAR, tilt sensor, UBEC); verify UART framing and 5V rail brownout protection on bench.
• **Phase 3–4 (Weeks 4–6):** Implement firmware tilt compensation ($d_{true} = d_{cal} \cdot \cos\theta$); assemble F450 quadcopter airframe; balance center of gravity; install silicone vibration isolation deck (>18 dB attenuation).
• **Phase 5–6 (Weeks 7–9):** Adapt PLY Forge WebGL engine for streaming UDP point cloud buffers; conduct tethered hover flight trials; scan facade with calibrated 2cm & 3.5cm simulated spalls.
• **Phase 7 (Week 10):** End-to-end faculty demonstration, automated PDF inspection report generation, and longitudinal deterioration baseline establishment.
• **Target Metrics:** Static distance MAE ≤ ±0.5cm; Tilt error residual ≤ ±1.0cm; Defect detection precision ≥ 90%, recall ≥ 85%; Continuous hover flight time ≥ 14 mins.

Component Group	Key Inclusions	Cost (INR)	Procurement Phasing
Sensors & Power Regulation	YDLIDAR X2 360° LiDAR (■7,189), DFRobot Tilt Sensor (■245), 5V 5A UBEC (■380)	**■7,814.00**	**Priority 1 (Immediate / Week 1–2)**
Airframe & Power Reservoir	F450 Frame Kit + Motors + ESCs (■5,200), 3S 4200mAh LiPo & B6 Charger (■3,800), Dampers (■740)	**■9,740.00**	**Priority 2 (Integration / Week 3–4)**
Avionics & Contingency	Pixhawk 2.4.8 Autopilot + GPS Kit (■6,850), Courier Shipping (■400), Risk Reserve (■2,000)	**■9,250.00**	**Priority 3 (Flight Trials / Week 5–6)**
TOTAL RECOMMENDED GRANT	**Complete Airborne Cyber-Physical SHM System (MVP Bench Option: ■8,214.00)**	**■27,124.00**	**Faculty Sign-Off Ready**

Project Leads: *Nisargkumar Joshi (25BCE0594, Ground Lead) & Priyam Sharma (25BAI0159, Aerial Lead)* | ***Guide Sign-off:*** *Dr. Senthil Kumar N (Associate Professor Grade 2, SCOPE)* |
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